

**ILLINOIS TECH** | College of Computing

**ITMM 485 / 585**  
**Dr. Gurram Gopal**

Legal and Ethical Issues in  
 Information Technology



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Ethical Concepts and Theories

**ILLINOIS TECH** | College of Computing

**Ch8: Intellectual Property Issues**  
**P4: IP Philosophical Foundations**  
**| Open-Source Software**



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**Learning Objectives:**

Upon completion of this lesson the students should be able to:

- Explain and discuss the philosophical and legal arguments used to protect intellectual property
- Describe the differences and similarities between the Free Software Foundation & the Open Source Initiative
- Explain how Creative Commons licensing of IP works
- Discuss the balance between public good and profit for IP rights holders

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**Philosophical Foundations for Intellectual Property**

- Three distinct types of traditional (philosophical) theories regarding property rights can be articulated, i.e., the:
  - 1) labor theory;
  - 2) utilitarian theory;
  - 3) personality theory.
- It is important to note that some theorists argue against property protection of any kind.
- Others argue against protecting intellectual property, but not against tangible property.

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**The Labor Theory of Property**

- Introduced by John Locke (17<sup>th</sup> century).
- According to the labor theory, a person is entitled to the results of his or her labor.
- Property rights are justified because of the one's physical "sweat of the brow."
- It makes sense for physical objects, but does it apply intellectual objects as well?

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## Criticisms of the Labor Theory

- The Labor Theory assumes that a property right is a *natural right* (but gives no clear defense of this view).
- It applies only to people who “own their bodies,” ruling out slaves and indigent peoples such as Native Americans.
- Intellectual works don’t always require the same kind of onerous labor.

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## The Utilitarian Theory of Property

- A property right is not a natural right; rather it is a conventional right granted by governments.
- Property rights should be granted because they provide an incentive to bring ideas into the marketplace (i.e. they have social utility).
- Utilitarian reasoning was used by the framers of the US Constitution (Article 1, Section 8).

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## Criticisms of the Utilitarian Theory of Property

- This theory (questionably) assumes that there must be an economic incentive to produce creative works; otherwise, creative works would not be produced.
- Utilitarian theory in general favors the interests of the greatest number of persons (affected by a social policy) at the expense of the minority population that is also affected by a social policy.

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## The Personality Theory of Property

- Traced back to G.W.F. Hegel (19<sup>th</sup> century).
- Property rights are not tied to labor or to economic incentives.
- A property right should be granted because of the personality of the author that is invested in the creative work (regardless of economic considerations).
- Consider that the commercial use of a creative work may not represent the author’s personality, as in the case of a Nike TV ad using John Lennon’s song “Revolution.”

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## Criticism of the Personality Theory

- The personality theory assumes that property rights are natural or moral rights.
- This theory ignores three factors:
  - a) the role of economic incentives;
  - b) the role of labor involved;
  - c) the fact that an author might not have invested her “true” personality in some creative work (such as a deliberate attempt on the part of the author of a creative work to deceive someone).

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## Summary of the Three Philosophical Theories of Property

|                           |                                                                                                                                                                                                                             |
|---------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>Labor Theory</i>       | Argues that a property right is a “natural right” and that property rights can be justified by the labor or “toil” that one invests in cultivating land or in creating a work of art.                                       |
| <i>Utilitarian Theory</i> | Argues that property rights are not natural rights but rather artificial rights created by the state. Property rights are granted to individuals and to corporations because they result in greater social utility overall. |
| <i>Personality Theory</i> | Argues that a property right is a “moral right” and that property rights are justified not because of labor or social utility but because creative works express the personalities of the authors that create them.         |

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## The Case Against Property Rights for Software

- Not everyone believes that property rights for software are justified.
- Some argue that while property rights for physical objects make sense, intellectual property rights for software does not.
- Richard Stallman has opposed copyright protection for software.

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## Alternative Frameworks to Property Rights for Software

- Stallman (2004) views software ownership as a form of “hoarding” that disregards the general welfare of society.
- He believes that software should be freely available for humankind rather than restricted by property rights.
- Stallman notes that the development of software in the computer industry has evolved from a spirit of cooperation and sharing to one in which cooperation is virtually forbidden.

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## Alternative Frameworks (Continued)

- Grodzinsky, Miller, and Wolf (2004) suggest that Stallman's position on why software should be free may have been influenced by the MIT culture of the 1970s, where source code could be freely exchanged.
- Although Stallman advocates for the view that software should be free, he intends “free” to refer to liberty not to price (or “free” as in free speech versus free beer).

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## Alternative Frameworks (Continued)

- In the late 1970s and early 1980s, the burgeoning computer industry hired many of the best software developers and programmers from academic computing labs.
- Some of those individuals took the software they developed with them, and some of that software eventually became proprietary.
- In response, Stallman began his GNU (Gnu's Not Unix) project in 1984, whose goal was to develop an entire Unix-like operating system that was “open” and freely accessible.

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## GNU and the Free Software Foundation (FSF)

- FSF was formed in 1985 to support of Stallman's GNU project.
- According to FSF, four “freedoms” are essential for free software, i.e., the *freedom to*:
  1. run the program, for any purpose;
  2. study how the program works, and adapt it for your needs;
  3. redistribute copies so you can help your neighbor;
  4. improve the program, and release your improvements to the public so that the whole community benefits.

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## The Open Source Initiative (OSI)

- OSI shares many of the same goals as FSF, including the ability of a software user to look at, understand, modify and redistribute the source code for that software.
- Like FSF, OSI requires that the source code for “open source software” (OSS) is freely available.
- So, both the OSS and FSF movements are similar with respect to their requirements for source code in the software development process.
- There are also important differences between OSS and FSF.

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## OSS and FSF (Continued)

- Raymond (2004) notes that OSS and FSF have different philosophies or “attitudes” because:
- FSF continues to focus on promoting its philosophical position that software should be free.
- OSS has concentrated its efforts more on promoting the open source model as an alternative methodology to “closed-source” development for software.
- OSS and FSF also differ with respect to requirements for how the software is used “downstream.”

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## OSS and FSF (Continued)

- FSF requires that all derivative pieces of software be subject to the original requirements and thus remain “open” and nonproprietary.
- OSS is more flexible with respect to its derivative software.
- FSF requires that users strictly adhere to its GPL (General Programming License) in all derivative uses of its software.
- OSS supports less restrictive licenses that permit programmers to alter the open source software and to release it as a proprietary product.

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## Alternative Frameworks and the Sharing of Information

- Stallman focuses his arguments specifically on why computer software, not necessarily all information, should be free (although some of his followers subscribe to the view: information wants to be free).
- But Stallman correctly recognizes that information is something that humans desire to *share* with one another.
- We do not need to accept his position on software being free to appreciate the force of Stallman’s insight about the broader notion of information.

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## Alternative Frameworks (continued)

- In order for information to be shared, it must be communicated.
- So, intellectual-property laws that prohibit or even discourage the communication of information undermine the very purpose of information as “something to be shared” (McFarland, 2004).

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## The “Common Good” Approach

- Stallman’s insight about the nature of information dovetails with the “common- good” approach to ethics.
- McFarland (2004, 2005) draws on some principles of virtue ethics and natural law theory in discussing how the “common good” applies to intellectual property issues.
- McFarland does not necessarily accept Stallman’s claim that software should be totally free.

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## “Information Wants to Be Shared” vs. “Information Wants to Be Free”

- Building on the insights of McFarland and others, we argue for the principle: *information wants to be shared.*
- This principle could be used as a starting point to guide policy debates.
- If we presume in favor of this principle, we can have fairer information policies.

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## Information Wants to Be Shared

- De George (2003) notes that original copyright laws (involving print media) were designed to encourage information distribution.
- With recent laws covering digital media, such as the DMCA and SBCTEA, the distribution of electronic information is now being inhibited, despite the fact that this kind of information exchange is easy and inexpensive.

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## Information Wants to Be Shared

- The original computing and Internet environments were governed by an implicit principle of *sharing* information.
- For example, Doug Englebart did not apply for a patent for the mouse that he invented.
- Also, Tim Berners-Lee did not copyright his HTML code that was eventually used as the standard protocol for the Web.

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## Preserving the Intellectual Commons

- We have framed laws and policies to protect the *physical commons* (i.e., parks, natural resources, etc.).
- The *intellectual commons* (of ideas) is now threatened by strong intellectual property laws.
- Many believe that we urgently need to act now to preserve the intellectual commons.

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## The Public Domain of Ideas

- The public domain of ideas is shrinking.
- Recall Eric Eldred's Web site of older books, which was forced to shut down.
- Books and information once easily available are now threatened as soon as they are converted into digital form.
- What will the future status of digital books be for interlibrary-loan practices?

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## The Environmental Movement: An Analogy for Cyberspace

- Boyle (2004) believes that a "political movement," similar to the environmental movement that emerged in the 1970s, is needed to preserve the public domain.
- Boyle points out that just as a political movement was necessary to save the environment from inevitable destruction, so too is an analogous movement needed to save the intellectual commons.

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## Environmental Analogy (Continued)

- Boyle notes that "the environment" almost "disappeared" under the simplistic claim of (highly individualistic) property rights.
- The public domain of information is disappearing because of similar interests.
- The environmental movement "invented" the concept of the environment so that farmers, consumers, hunters, and bird watchers could "discover themselves as environmentalists."

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### Environmental Analogy (Continued)

- Boyle concludes that we may also need to invent (or perhaps “reinvent”) the concept of the public domain in order to call into being a coalition of stakeholders that could protect it.
- He suggests that it might still be possible to reclaim the public domain of ideas, if we act responsibly.

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### Defending the Principle: *Information Wants to Be Shared*

- If we defend the principle that information wants to be shared (but not totally free), then it will be possible to frame reasonable intellectual property policies that both:
  - a) encourage the flow of information in digital form;
  - b) reward fairly the creators of intellectual objects, including software manufacturers.
- One promising scheme for accomplishing these objectives is the Creative Commons initiative.

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